OOP-TASK

Lab-06

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Lab Tasks:

Q1)

1.cpp

```cpp
#include <iostream>
using namespace std;
class Person{
    private:
    string name;
    int age;
    public:
    Person(string name,int age){
       this->name = name;
       this->age = age;
    }
    void displayinfo(){
        cout << "Name :" << name << endl;
        cout << "Age :" << age << endl;
    }
};
class Student: public Person{
    private:
    string rollNo;
    public:
    Student(string name, int age, string rol):Person(name, age), rollNo(rol){}
    void displayStudent(){
        displayinfo();
        cout << "Roll no :" << rollNo << endl;
    }

};
int main(){
    Student s1("Abdullah",18,"25k-0868");
    s1.displayStudent();
}
```

Output:

```
C:\Users\marti\OneDrive\Documents\1.exe
Name :Abdullah
Age :18
Roll no :25k-0868

--------------------------------
Process exited after 1.038 seconds with return value 0
Press any key to continue . . .
```

Q2):

[*] 1.cpp

```cpp
#include <iostream>
using namespace std;
class Employee{
    private:
    int ID;
    double salary;
    public:
    Employee(double salary,int Id ):salary(salary),ID(Id){}
    double getSalary(){
        return salary;
    }
    void showEmployee(){
        cout << "ID:" << ID << endl;
        cout << "Salary:" << salary << endl;
    }
};
class Manager: public Employee{
    private:
    double bonus;
    public:
    Manager(int id,double salary,double bonus):Employee(salary,id),bonus(bonus){}
    void showManager(){
        showEmployee();
        cout << "Bonus:" << bonus << endl;
        cout << "Complete Salary:" << getSalary() + bonus << endl;
    }
};
int main(){
    Manager m1(101, 75000, 12000);
    m1.showManager();
    return 0;
}
```

```
C:\Users\marti\OneDrive\Documents\1.exe
ID:101
Salary:75000
Bonus:12000
Complete Salary:87000

--------------------------------
Process exited after 1.126 seconds with return value 0
Press any key to continue . . .
```

Q3):

1.cpp

```cpp
#include <iostream>
using namespace std;
class Writer{
    public:
    void write(){
        cout << "Writer is writing..." << endl;
    }
};
class Speaker{
    public:
    void speak(){
        cout << "Speaker is speaking..." << endl;
    }
};
class Author: public Writer, public Speaker{
    public:
    void showDetails(){
        cout << "Author Details:" << endl;
        write();
        speak();
    }
};
int main(){
    Author a1;
    a1.showDetails();
    return 0;
}
```

```
C:\Users\marti\OneDrive\Documents\1.exe
Author Details:
Writer is writing...
Speaker is speaking...

--------------------------------
Process exited after 0.7109 seconds with return value 0
Press any key to continue . . .
```

Q4):

```cpp
class Person{
    private:
    string name;
    int age;
    public:
    Person(string name,int age){
        this->name = name;
        this->age = age;
    }
    void displayinfo(){
        cout << "Name :" << name << endl;
        cout << "Age :" << age << endl;
    }
};
class Employee: public Person{
    private:
    double salary;
    public:
    Employee(string name,int age,double salary):Person(name,age),salary(salary){}
    void displayEmployee(){
        displayinfo();
        cout << "Salary :" << salary << endl;
    }
};
class Developer: public Employee{
    private:
    string programmingLanguage;
    public:
    Developer(string name,int age,double salary,string lang):Employee(name,age,salary),programmingLanguage(lang){}
    void displayDeveloper(){
        displayEmployee();
        cout << "Programming Language :" << programmingLanguage << endl;
    }
};
int main(){
    Developer d1("Abdullah",19,5000,"Python");
    d1.displayDeveloper();
    return 0;
}
```

C:\Users\marti\OneDrive\Documents\1.exe

```
Name :Abdullah
Age :19
Salary :5000
Programming Language :Python

--------------------------------
Process exited after 1.068 seconds with return value 0
Press any key to continue . . .
```

Q5):

[*] 1.cpp

```cpp
#include <iostream>
using namespace std;
class Account{
    private:
    string accountNumber;
    public:
    Account(string accNum):accountNumber(accNum){}
    void showAccount(){
        cout << "Account Number: " << accountNumber << endl;
    }
};
class SavingsAccount : public Account{
    private:
    double interestRate;
    public:
    SavingsAccount(string accNum, double rate):Account(accNum),interestRate(rate){}
    void calculateInterest(){
        cout << "Calculating interest at rate: " << interestRate << "%" << endl;
    }
};
class CurrentAccount : public Account{
    private:
    double overdraftLimit;
    public:
    CurrentAccount(string accNum, double limit):Account(accNum),overdraftLimit(limit){}
    void calculateOverdraft(){
        cout << "Calculating overdraft with limit: " << overdraftLimit << endl;
    }
};
int main(){
    SavingsAccount sa1("SA12345", 5.0);
    CurrentAccount ca1("CA54321", 1000.0);
    cout << "Savings Account Details:" << endl;
    sa1.showAccount();
    sa1.calculateInterest();
    cout << "\nCurrent Account Details:" << endl;
    ca1.showAccount();
    ca1.calculateOverdraft();
    return 0;
}
```

C:\Users\marti\OneDrive\Documents\1.exe

```
Savings Account Details:
Account Number: SA12345
Calculating interest at rate: 5%

Current Account Details:
Account Number: CA54321
Calculating overdraft with limit: 1000

--------------------------------
Process exited after 1.26 seconds with return value 0
Press any key to continue . . .
```

Q6):

```cpp
class University{
   private:
   string name;
   public:
   University(string name){
    this->name = name;
   }
    void showUniversity(){
     cout << "University Name: " << name << endl;
    }
};
class Faculty: public University{
    private:
    string department;
    public:
    Faculty(string name, string dept):University(name),department(dept){}
    void showFaculty(){
        showUniversity();
        cout << "Department: " << department << endl;
    }
};
class Administration{
    private:
    string adminName;
    public:
    Administration(string adminName){
        this->adminName = adminName;
    }
    void showAdministration(){
        cout << "Administrator: " << adminName << endl;
    }
};
class HOD: public Faculty, public Administration{
    public:
    HOD(string uniName, string dept, string adminName):Faculty(uniName, dept),Administration(adminName){}
    void showHODDetails(){
        showFaculty();
        showAdministration();
    }
```

```
int main(){
    HOD hod1("FAST University", "Computer Science", "Abdul Saeed");
    hod1.showHODDetails();
    return 0;
}
```

```
C:\Users\marti\OneDrive\Documents\1.exe
University Name: FAST University
Department: Computer Science
Administrator: Abdul Saeed

--------------------------------
Process exited after 1.385 seconds with return value 0
Press any key to continue . . .
```

Q7):

```cpp
#include <iostream>
using namespace std;
class Base {
public:
    int publicVar;
protected:
    int protectedVar;
private:
    int privateVar;
public:
    Base(int pub, int prot, int priv) : publicVar(pub), protectedVar(prot), privateVar(priv) {}
    void showBaseDetails() {
        cout << "Base Class Details:" << endl;
        cout << "Public Variable: " << publicVar << endl;
        cout << "Protected Variable: " << protectedVar << endl;
        cout << "Private Variable: " << privateVar << endl;
    }
};
class PublicDerived : public Base {
public:
    PublicDerived(int pub, int prot, int priv) : Base(pub, prot, priv) {}
    void showPublicDerivedDetails() {
        cout << "Public Derived Class Details:" << endl;
        cout << "Public Variable: " << publicVar << endl; // Accessible
        cout << "Protected Variable: " << protectedVar << endl; // Accessible
        // cout << "Private Variable: " << privateVar << endl; // Not Accessible
    }
};
class ProtectedDerived : protected Base {
public:
    ProtectedDerived(int pub, int prot, int priv) : Base(pub, prot, priv) {}
    void showProtectedDerivedDetails() {
        cout << "Protected Derived Class Details:" << endl;
        cout << "Public Variable: " << publicVar << endl; // Accessible
        cout << "Protected Variable: " << protectedVar << endl; // Accessible
        // cout << "Private Variable: " << privateVar << endl; // Not Accessible
    }
};
class PrivateDerived : private Base {
```

```
};
class PrivateDerived : private Base {
public:
    PrivateDerived(int pub, int prot, int priv) : Base(pub, prot, priv) {}
    void showPrivateDerivedDetails() {
        cout << "Private Derived Class Details:" << endl;
        cout << "Public Variable: " << publicVar << endl; // Accessible
        cout << "Protected Variable: " << protectedVar << endl; // Accessible
        // cout << "Private Variable: " << privateVar << endl; // Not Accessible
    }
};
int main() {
    Base baseObj(1, 2, 3);
    PublicDerived pubDerivedObj(4, 5, 6);
    ProtectedDerived protDerivedObj(7, 8, 9);
    PrivateDerived privDerivedObj(10, 11, 12);

    cout << "Accessing Base Class from Main:" << endl;
    cout << "Public Variable: " << baseObj.publicVar << endl; // Accessible
    // cout << "Protected Variable: " << baseObj.protectedVar << endl; // Not Accessible
    // cout << "Private Variable: " << baseObj.privateVar << endl; // Not Accessible

    cout << "\nAccessing Public Derived Class from Main:" << endl;
    pubDerivedObj.showPublicDerivedDetails();

    cout << "\nAccessing Protected Derived Class from Main:" << endl;
    protDerivedObj.showProtectedDerivedDetails();

    cout << "\nAccessing Private Derived Class from Main:" << endl;
    privDerivedObj.showPrivateDerivedDetails();

    return 0;
}
```

C:\Users\marti\OneDrive\Documents\1.exe

```
Accessing Base Class from Main:
Public Variable: 1

Accessing Public Derived Class from Main:
Public Derived Class Details:
Public Variable: 4
Protected Variable: 5

Accessing Protected Derived Class from Main:
Protected Derived Class Details:
Public Variable: 7
Protected Variable: 8

Accessing Private Derived Class from Main:
Private Derived Class Details:
Public Variable: 10
Protected Variable: 11

--------------------------------
Process exited after 1.855 seconds with return value 0
Press any key to continue . . .
```

Q8):

```cpp
#include <iostream>
using namespace std;
class person {
public:
    void show() {
        cout << "Person" << endl;
    }
};
class student : public person {};
class employee : public person {};
class teachingAssistant : public student, public employee {};
int main() {
    teachingAssistant obj;
    // obj.show(); // This will cause a compilation error due to ambiguity
    obj.student::show(); // This resolves the ambiguity by specifying which base class's show() to call
    return 0;
}
```

```
C:\Users\marti\OneDrive\Documents\1.exe
Person

--------------------------------
Process exited after 4.077 seconds with return value 0
Press any key to continue . . .
```

Q9):

```cpp
#include <iostream>
using namespace std;
class Person {
public:
    void show() {
        cout << "Person" << endl;
    }
};
class Student : virtual public Person {};
class Employee : virtual public Person {};
class TeachingAssistant : public Student, public Employee {};
int main() {
    TeachingAssistant obj;
    obj.show(); // No ambiguity
    cout << "Memory address of Person from Student path: " << static_cast<Student*>(&obj) << endl;
    cout << "Memory address of Person from Employee path: " << static_cast<Employee*>(&obj) << endl;
    return 0;
}
```

C:\Users\marti\OneDrive\Documents\1.exe

```
Person
Memory address of Person from Student path: 0x6ffe10
Memory address of Person from Employee path: 0x6ffe18

--------------------------------
Process exited after 5.352 seconds with return value 0
Press any key to continue . . .
```

HOME TASKS:

Task1:

```cpp
#include <iostream>
using namespace std;
class Vehicle{
    private:
    string brand;
    int speed;
    public:
    Vehicle(string brand,int speed):brand(brand),speed(speed){}
    void showDetail(){
        cout << "Brand:" << brand << endl;
        cout << "speed:" << speed << endl;
    }
};
class Car : public Vehicle{
    private:
    string fuelType;
    public:
    Car(string fuel,string brand,int speed):fuelType(fuel),Vehicle(brand,speed){}
    void showCardetails(){
        showDetail();
        cout << "Fuel Type:" << fuelType << endl;
    }
};
int main(){
    Car c1("Patrol","toyota",25);
    c1.showCardetails();
}
```

C:\Users\marti\OneDrive\Documents\1.exe

```
Brand:toyota
speed:25
Fuel Type:Patrol

--------------------------------
Process exited after 1.51 seconds with return value 0
Press any key to continue . . .
```

Task2:

```cpp
#include <iostream>
using namespace std;
class Shape{
    protected:
    double radius;
    public:
    Shape(double r):radius(r){}
};
class Circle: public Shape{
    public:
    Circle(double r):Shape(r){}
    double area(){
        return 3.14 * radius * radius;
    }
};
int main(){
    Circle c1(5.0);
    cout << "Area of Circle: " << c1.area() << endl;
    return 0;
}
```

C:\Users\marti\OneDrive\Documents\1.exe

```
Area of Circle: 78.5

--------------------------------
Process exited after 1.145 seconds with return value 0
Press any key to continue . . .
```

Task3:

```cpp
#include <iostream>
using namespace std;
class Driver {
public:
    void drive() {
        cout << "Driver is driving..." << endl;
    }
};
class Mechanic {
public:
    void repair() {
        cout << "Mechanic is repairing..." << endl;
    }
};
class Technician : public Driver, public Mechanic {
public:
    void showDetails() {
        cout << "Technician Details:" << endl;
        drive();
        repair();
    }
};
int main() {
    Technician t1;
    t1.showDetails();
    return 0;
}
```

C:\Users\marti\OneDrive\Documents\1.exe

```
Technician Details:
Driver is driving...
Mechanic is repairing...

--------------------------------
Process exited after 1.502 seconds with return value 0
Press any key to continue . . .
```

Task4:

```cpp
#include <iostream>
using namespace std;
class Appliance {
    private:
    string brand;
    int power;
    public:
    Appliance(string brand,int power):brand(brand),power(power){}
    void showDetails(){
        cout << "Brand:" << brand << endl;
        cout << "Power:" << power << "W" << endl;
    }
};
class WashingMachine : public Appliance {
    private:
    int capacity;
    public:
    WashingMachine(string brand,int power,int capacity):Appliance(brand,power),capacity(capacity){}
    void showWashingMachineDetails(){
        showDetails();
        cout << "Capacity:" << capacity << "kg" << endl;
    }
};
class SmartWashingMachine : public WashingMachine {
    private:
    string smartControl;
    public:
    SmartWashingMachine(string brand,int power,int capacity,string smartControl):WashingMachine(brand,power,capacity),smartControl(smartControl){}
    void showSmartWashingMachineDetails(){
        showWashingMachineDetails();
        cout << "Smart Control:" << smartControl << endl;
    }
};
int main(){
    SmartWashingMachine swm1("LG",2000,7,"Wi-Fi");
    swm1.showSmartWashingMachineDetails();
    return 0;
}
```

C:\Users\marti\OneDrive\Documents\1.exe

```
Brand:LG
Power:2000W
Capacity:7kg
Smart Control:Wi-Fi

--------------------------------
Process exited after 1.518 seconds with return value 0
Press any key to continue . . .
```

Task5:

```cpp
#include <iostream>
using namespace std;
class Media{
    private:
    string title;
    public:
    Media(string t):title(t){}
    void showTitle(){
        cout << "Title: " << title << endl;
    }
};
class Book:public Media{
    private:
    string author;
    public:
    Book(string t,string a):Media(t),author(a){}
    void showBookDetails(){
        showTitle();
        cout << "Author: " << author << endl;
    }
};
class Vedio:public Media{
    private:
    int duration;
    public:
    Vedio(string t,int d):Media(t),duration(d){}
    void showVedioDetails(){
        showTitle();
        cout << "Duration: " << duration << " minutes" << endl;
    }
};
class Audio:public Media{
    private:
    string artist;
    public:
    Audio(string t,string a):Media(t),artist(a){}
    void showAudioDetails(){
        showTitle();
```

```
class Vedio:public Media{
    private:
    int duration;
    public:
    Vedio(string t,int d):Media(t),duration(d){}
    void showVedioDetails(){
        showTitle();
        cout << "Duration: " << duration << " minutes" << endl;
    }
};
class Audio:public Media{
    private:
    string artist;
    public:
    Audio(string t,string a):Media(t),artist(a){}
    void showAudioDetails(){
        showTitle();
        cout << "Artist: " << artist << endl;
    }
};
int main(){
    Book b1("Intersatller","Christopher Nolan");
    Vedio v1("Vedio",148);
    Audio a1("Lecture","Miss Hijaab");

    cout << "Book Details:" << endl;
    b1.showBookDetails();

    cout << "\nVedio Details:" << endl;
    v1.showVedioDetails();

    cout << "\nAudio Details:" << endl;
    a1.showAudioDetails();

    return 0;

}
```

C:\Users\marti\OneDrive\Documents\1.exe

```
Book Details:
Title: Intersatller
Author: Christopher Nolan

Vedio Details:
Title: Vedio
Duration: 148 minutes

Audio Details:
Title: Lecture
Artist: Miss Hijaab

--------------------------------
Process exited after 1.192 seconds with return value 0
Press any key to continue . . .
```

Task6:

```cpp
#include <iostream>
using namespace std;
class Device{
    private:
    string model;
    public:
    Device(string model):model(model){}
    void ShowModel(){
        cout << "Model" << model << endl;
    }
};
class Computer: public Device{
    private:
    string processor;
    public:
    Computer(string model,string proc):Device(model),processor(proc){}
    void ShowComputerDetails(){
        ShowModel();
        cout << "Processor:" << processor << endl;
    }
};
class Network{
    private:
    string networkType;
    public:
    Network(string type):networkType(type){}
    void ShowNetwork(){
        cout << "Network Type:" << networkType << endl;
    }
};
class Server: public Computer, public Network{
    public:
    Server(string model,string proc,string netType):Computer(model,proc),Network(netType){}
    void ShowServerDetails(){
        ShowComputerDetails();
        ShowNetwork();
    }
};
int main(){
    Server s1("Dell PowerEdge","Intel Xeon","Ethernet");
    s1.ShowServerDetails();
    return 0;
```

C:\Users\marti\OneDrive\Documents\1.exe

```
ModelDell PowerEdge
Processor:Intel Xeon
Network Type:Ethernet

--------------------------------
Process exited after 1.545 seconds with return value 0
Press any key to continue . . .
```

Task7:

```cpp
#include <iostream>
using namespace std;
class BankAccount
{
private:
    int accountNumber;
    double balance;
public:
    BankAccount(int accNum, double bal) : accountNumber(accNum), balance(bal) {}
    void displayAccountInfo() const {
        cout << "Account Number: " << accountNumber << endl;
        cout << "Balance: $" << balance << endl;
    }
};
class SavingsAccount : private BankAccount
{public:
    SavingsAccount(int accNum, double bal) : BankAccount(accNum, bal) {}
    void showSavingsInfo() const {
        displayAccountInfo(); // Accessing base class function
    }
};
int main() {
    SavingsAccount mySavings(12345, 1000.0);
    // mySavings.displayAccountInfo(); // This will cause a compilation error
    mySavings.showSavingsInfo(); // This works because it's a member function of SavingsAccount
    return 0;
}
```

C:\Users\marti\OneDrive\Documents\1.exe

```
Account Number: 12345
Balance: $1000

--------------------------------
Process exited after 1.183 seconds with return value 0
Press any key to continue . . .
```

Task8:

```cpp
#include <iostream>
using namespace std;
class Machine {
public:
    void show() {
        cout << "Machine" << endl;
    }
};
class Printer : public Machine {};
class Scanner : public Machine {};
class AllInOne : public Printer, public Scanner {};
int main() {
    AllInOne obj;
    // obj.show(); // This will cause a compilation error due to ambiguity
    obj.Printer::show(); // This resolves the ambiguity by specifying which base class's show() to call
    return 0;
}
```

```
C:\Users\marti\OneDrive\Documents\1.exe
Machine

--------------------------------
Process exited after 1.471 seconds with return value 0
Press any key to continue . . .
```

Task9:

```cpp
#include <iostream>
using namespace std;
class Device {
public:
    void show() {
        cout << "Device" << endl;
    }
};
class Camera : virtual public Device {};
class Phone : virtual public Device {};
class SmartPhone : public Camera, public Phone {};
int main() {
    SmartPhone obj;
    obj.show(); // No ambiguity
    cout << "Memory address of Device from Camera path: " << static_cast<Camera*>(&obj) << endl;
    cout << "Memory address of Device from Phone path: " << static_cast<Phone*>(&obj) << endl;
    return 0;
}
```

```
C:\Users\marti\OneDrive\Documents\1.exe
Device
Memory address of Device from Camera path: 0x6ffe10
Memory address of Device from Phone path: 0x6ffe18

--------------------------------
Process exited after 1.076 seconds with return value 0
Press any key to continue . . .
```

FINAL LAB TASK:(Multiple inheritance in student management system)

```
#include <iostream>
using namespace std;
class Person {
protected:
    string name;
public:
    void setName(string n) {
        name = n;
    }
    void show() {
        cout << "Name: " << name << endl;
    }
};
class Student : virtual public Person {
private:
    int studentID;
public:
    void setStudentID(int id) {
        studentID = id;
    }
    void show() {
        Person::show();
        cout << "Student ID: " << studentID << endl;
    }
};
class Employee : virtual public Person {
private:
    int employeeID;
public:
    void setEmployeeID(int id) {
        employeeID = id;
    }
    void show() {
        Person::show();
        cout << "Employee ID: " << employeeID << endl;
    }
};
```

```
};
class Employee : virtual public Person {
private:
    int employeeID;
public:
    void setEmployeeID(int id) {
        employeeID = id;
    }
    void show() {
        Person::show();
        cout << "Employee ID: " << employeeID << endl;
    }
};
class TeachingAssistant : public Student, public Employee {
public:
    void show() {
        Student::show();
        Employee::show();
    }
};
int main() {
    TeachingAssistant ta;
    ta.setName("Abdullah");
    ta.setStudentID(02120);
    ta.setEmployeeID(12345);
    ta.show();
    return 0;
}
```

C:\Users\marti\OneDrive\Documents\1.exe

```
Name: Abdullah
Student ID: 1104
Name: Abdullah
Employee ID: 12345

--------------------------------
Process exited after 1.077 seconds with return value 0
Press any key to continue . . .
```